

Language Models That Accurately Represent Syntactic Structure Exhibit Higher Representational Similarity To The Brain

Anonymous CogSci submission

Abstract

Transformers have demonstrated exceptional alignment to the brain during language processing. However, it is still debated what linguistic features drive this alignment. Here, we investigated how the accurate representation of syntactic dependencies in transformers influences brain alignment. We used the fMRI data from the Mother Of all Unification Studies in which participants read Dutch sentences with varying syntactic complexity. The same sentences were fed through monolingual and multilingual transformer models. Next, representational similarity analysis was performed between the transformer embeddings and the fMRI data extracted from the Left posterior Medial Temporal Gyrus, a region associated with syntax processing. By establishing the accuracy of the dependency representations in the transformers with DepProbe, a labelled structural probe, we could then compare the representation similarity scores with the accuracy of the dependency representations across layers. Our analysis revealed four key findings: 1) Models that accurately represent dependency information show greater similarity to the brain, 2) Monolingual models and multilingual models do not differ significantly in their brain similarity or accuracy of dependency representations, 3) The relationship between brain similarity and accuracy of dependency representations holds, regardless of the syntactic complexity of a sentence, and 4) Higher syntactic complexity leads to lower accuracies of dependency representations but higher brain similarity. Together, these results show that syntactic dependencies play an important role in driving brain similarity.

Keywords: Large Language Models; Dependency Structure; Syntactic probing; Representational Similarity Analysis

Introduction

In recent years, transformers have become the ubiquitous architecture for state-of-the-art language models (Vaswani et al., 2017). While the linguistic performance of these large language models (LLMs) is unrivalled (Qiu et al., 2020), it is still debated to which extent the representations that drive this performance are analogous to the human brain. Recent studies have shown that the representations of transformer models overlap partially with the human brain during the reading of isolated sentences (Caucheteux & King, 2022, 2020; Schrimpf et al., 2021), natural listening (Caucheteux, Gramfort, & King, 2021; Schrimpf et al., 2021), and natural conversation (Cai, Hadjinicolaou, Paulk, Williams, & Cash, 2023). However, there is still an ongoing debate on how linguistic features are represented in LLMs, and how this relates to their ability to predict—or correlate to, the human brain.

There are very few papers that have investigated the influence of representing dependency information on the brain

alignment of language models. Previous research by Murphy, Talukdar, and Mitchell (2012) has shown that including dependency information in semantic models increases the models' ability to predict brain activation. In this study, the authors extracted semantic features of words from a corpus based on word co-occurrence and various linguistic properties. For example, the parts-of-speech model produced features that included the basic syntactic function of a word. These features were then combined in a regularised manner to build a linear model that predicts individual voxel activations of the cortex during a property generation task. Their findings indicate that the semantic models that incorporate the dependency relations between words are best suited to predict brain activation. This result has been corroborated by Abnar, Ahmed, Mijnheer, and Zuidema (2017), who compared the brain predictivity of different word embedding models. They found that for predicting voxel activations from word embeddings, the word2vec model that included information about dependency structures outperformed other embedding models. These results substantiate the notion that including information about dependency structures in language models increases their similarity to the brain.

Note that the two papers mentioned above were investigating previous-generation language models. Research directly investigating how the representation of dependency structures influences the brain alignment of transformers is scarce. However, the paper by Oota, Gupta, and Toneva (2023) has shown that the removing syntactic information related to dependencies from transformers leads to a larger drop in brain alignment than the removal of surface or semantic information.

Contrastingly, the recent findings by Kauf, Tuckute, Levy, Andreas, and Fedorenko (2023) have indicated that semantic content is the most important driver of brain similarity in transformers. Moreover, the study by Gauthier and Levy (2019) even demonstrated that language models exhibited enhanced predictive capabilities for brain activity under conditions where syntactic information within these models was abolished. In short, the extent to which syntactic dependencies influence brain alignment is unclear.

The Present Study

The present study explores the relationship between the ability of transformers to accurately represent dependency struc-

tures and their representational similarity (RS) to the brain. Focusing on five pretrained transformer models, including both Dutch monolingual (BERTje, RobBERT) and multilingual (MBERT, XLM-Roberta, XLM-v) variants, we assess their dependency representations using DepProbe (Müller-Eberstein, van der Goot, & Plank, 2022). This tool allows us to examine how well these models construct dependency trees, as measured by the Labelled Attachment Score (LAS), a metric indicating the accuracy of dependency parsing.

We investigate the RS of transformer models by comparing their embeddings with brain data collected from participants performing a reading task in Dutch. We are particularly interested in the brain data from the Left posterior Middle Temporal Gyrus (LpMTG), a brain area associated with syntax processing (Hagoort & Indefrey, 2014; Tyler, Cheung, Devreux, & Clarke, 2013).

Firstly, our study hypothesizes that transformers with higher LAS scores, indicating a more accurate representation of dependency structures, will show greater RS with brain data. To control for the importance of word meaning for representational similarity, we compare the results of the RSA with a control condition in which the word order of the sentences was perturbed, effectively eliminating syntactic structure while keeping the meaning of individual words intact.

How well LLMs represent syntactic dependencies may differ from language to language. The study by Pérez-Mayos, Táboas García, Mille, and Wanner (2021) found that multilingual models such as MBERT and XLM-R outperform monolingual models such as BETO in Spanish, but, in turn, are outperformed by the monolingual BERT in English. The authors suggest that this difference may be due to the fact that, in contrast to English, Spanish is a morphologically rich language and thus may explicitly show syntactic agreement through morphology. In this regard, Dutch is closer to English and we may expect a similar pattern. Thus, secondly, we expect Dutch monolingual models to outperform multilingual models in both LAS and RS.

Thirdly, we consider the role of syntactic complexity, as measured by left-branching complexity (LBC), in influencing RS, LAS, and their relationship. This approach aims to deepen our understanding of the similarity between the human language system and LLMs.

Materials and Methods

fMRI data

The fMRI data used in the present study was obtained from the Mother Of all Unification Studies (MOUS) (Schoffelen et al., 2019). In the MOUS dataset, the brain signals of 204 participants were recorded using fMRI and MEG while they read or listened to Dutch sentences or word lists. Here we only used the fMRI data in which the participants read the sentences and, thus, all references to the stimuli will from this point forward denote the textual stimuli. The participants in this study were presented with one of five different scenarios denoting different subsets of the stimuli, each containing

60 sentences and 60 word sequences. We chose to only analyze the participants who were presented with the stimuli from scenario 5. This scenario was chosen because no problems during the data collection were reported by Schoffelen and colleagues. However, the events log of one participant revealed that they were not presented with all of the stimuli present in scenario five and, thus, this participant was excluded. Data from the remaining participants was used for further analysis ($n = 16$).

Syntactic Complexity The sentences presented to the participants in the MOUS were either syntactically complex, containing a relative clause, or syntactically simpler, containing a main clause and a simple subordinate clause. The study by Uddén et al. (2022) that used the same dataset utilized a particular measure of syntactic complexity: left-branching complexity (LBC). For each sentence, when parsing from left to right, this value is equal to the maximum number of dependents that are at the point of reading not yet assigned to a verb, see Figure 1. The manually annotated sentences and their corresponding LBC values were provided graciously by the authors of Uddén et al. (2022).

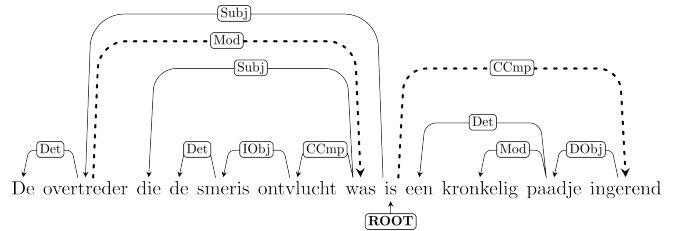


Figure 1: Example of a complex sentence from the MOUS. If we parse this sentence from left to right, we see that "overtreder" is dependent on the verb "is", "die" is dependent on the verb "was", "smeris" is dependent on the verb "ontvlucht", and "ontvlucht" is the dependent on the verb "was". Thus, this makes the maximum number of open verbal dependencies 4 during the the retrieval of the word "ontvlucht". Reproduced with permission from Harbusch et al. (2015).

Transformers

Five different transformer models were selected to model sentence processing. Crucially, two of these models were monolingual: BERTje (de Vries et al., 2019), and RobBERT (Delobelle, Winters, & Berendt, 2020), and three were multilingual: MBERT (Devlin, Chang, Lee, & Toutanova, 2019), XLM-r (Conneau et al., 2019), and XLM-v (Liang et al., 2023). All sentence embeddings were generated using the pre-trained versions of these models without finetuning. Note that, while the architecture for these models is largely identical, some key differences remain. For instance, mBERT is pre-trained using only data from Wikipedia, while the four other models were pre-trained on more diverse sources. An overview of the overlap and differences between the models can be found in Table 1.

Short name	Checkpoint	#lgs	L	HS	FF	A	V	P	Data	Dutch size	Masking
BERTje	GroNLP / bert-base-dutch-cased	1	12	768	3072	12	30k	110M	Different sources	12 GB	Static
mBERT	bert-base-multilingual-cased	104	12	768	3072	12	110k	110M	Wikipedia dump	Unknown	Static
RobBERT	pdelobelle/robbert-v2-dutch-base	1	12	768	3072	12	40k	117M	OSCAR corpus	39 GB	Dynamic
XLm-v	facebook/xlm-v-base	100	12	768	3072	12	1M	270M	CC-100 corpus	29.3 GB	Dynamic
XLm-r	xlm-roberta-base	100	12	768	3072	12	250K	270M	CC-100 corpus	29.3 GB	Dynamic

Table 1: Overview of the transformer models used in the present study

DepProbe

To extract the dependency representations from transformers, we used DepProbe¹ (Müller-Eberstein et al., 2022). By combining the structural probe by Hewitt and Manning (2019) with relational probing, DepProbe allows us to extract fully directed and labelled dependency trees from transformer models. Specifically, one can feed a set of sentences to DepProbe and choose from which two layers the contextualized word embeddings should be extracted. One of these layers is used to extract structural information about the sentence from the embeddings, while the other is used to extract relational information. More precisely, DepProbe applies a linear transformation $B \in \mathbb{R}^{e \times b}$ to the contextual embeddings. This maps the embeddings into a subspace where the distance measure

$$d_B(\mathbf{h}_i, \mathbf{h}_j) = \sqrt{(\mathbf{B}\mathbf{h}_i - \mathbf{B}\mathbf{h}_j)^T (\mathbf{B}\mathbf{h}_i - \mathbf{B}\mathbf{h}_j)} \quad (1)$$

between $\mathbf{h}_i$ and $\mathbf{h}_j$ can be optimized to approach the distance between two words $d_P(w_i, w_j)$ through gradient descent. The loss is then defined as the mean absolute difference between all word pairs of a sentence:

$$\mathcal{L}_B(s) = \frac{1}{N^2} \sum_{i=0}^N \sum_{j=0}^N |d_P(w_i, w_j) - d_B(\mathbf{h}_i, \mathbf{h}_j)|. \quad (2)$$

The undirected dependency graph can be extracted by applying the minimum spanning tree algorithm (Jarník, 1930; Prim, 1957) to the distances of word pairs in the sentences.

On top of extracting the undirected dependency graph, DepProbe predicts the labels that govern the dependencies. For each edge connected to a word w_i , a single label describes this relation. Thus, the label classification task with l relations can be defined as a linear transformation $L \in \mathbb{R}^{e \times l}$ where the probability that the connection of a word r_i is of label l_k can be given by:

$$p(r_i = l_k | w_i) = \text{softmax}(\mathbf{L}\mathbf{h}_i)_k \quad (3)$$

where the loss, given the gold label r_i^* for each word w_i , can be defined as:

$$\mathcal{L}_L(s) = -\frac{1}{N} \sum_{i=0}^N \ln p(r_i^* | w_i) \quad (4)$$

The direction of a connection is then determined by looking at the location of the root, thus making the head of the connection between the word w_i and word w_j simply the word with

the lowest number of edges to the root. This allows DepProbe to use fewer parameters than previous methods, therefore enabling a more direct investigation into the dependency representations of transformers. This complete process enables DepProbe to extract fully labelled and directed dependency trees that can be evaluated through the metric called labelled attachment score (LAS), the percentage of correctly placed and labelled directed edges in a syntax tree.

Procedure

A schematic overview of the procedure can be found in Figure 2. To use DepProbe on Dutch sentences, we first needed to train it on Dutch dependency data. Therefore, we obtained the annotated sentences of the Alpino treebank (van der Beek, Bouma, Malouf, & van Noord, 2001) from the Universal Dependencies (UD) database (Nivre et al., 2020). Then, for each of the five transformer models, the annotated sentences were fed through the model and contextualised embeddings were extracted from two layers, one to extract the structural information and the other to extract the relational information. Because earlier research has shown that brain-like representations of transformer models are mostly found in the middle layers (Schrimpf et al., 2021; Kumar et al., 2022; Goldstein et al., 2023), the structural and relational layers were chosen from layers 4 to 11. For each transformer model, DepProbe was trained using the embeddings from every possible combination of layers without hyperparameter tuning. These embeddings were then used to predict the dependency structure and labels for each of the sentences present in the Alpino dataset. The performance of DepProbe was evaluated on the LAS. The results of the training procedure followed our expectations, with the LAS peaking in the middle to late layers. In addition, the results indicated that changing the relational layer had relatively little effect on the resulting LAS compared to the structural layer, thus hereafter all results from DepProbe are averaged over the relational layers.

Subsequently, the DepProbe models were fed the sentences from the MOUS. We used a state-of-the-art neural parser, the biaffine parser (Dozat & Manning, 2016) to deliver the gold parse for the MOUS sentences. We then evaluated the predictions of the DepProbe models on each of the measures described above using the parses of the biaffine model.

The fMRI data used in the present study were obtained from the MOUS (Schoffelen et al., 2019). After preprocessing in fMRIPrep (Esteban et al., 2019) ROI extraction of the LpMTG was performed in FSL (Jenkinson, Beckmann, Behrens, Woolrich, & Smith, 2012) by applying a 10mm spherical mask using the MNI coordinates reported in (Uddén

¹<https://github.com/personads/depprobe>

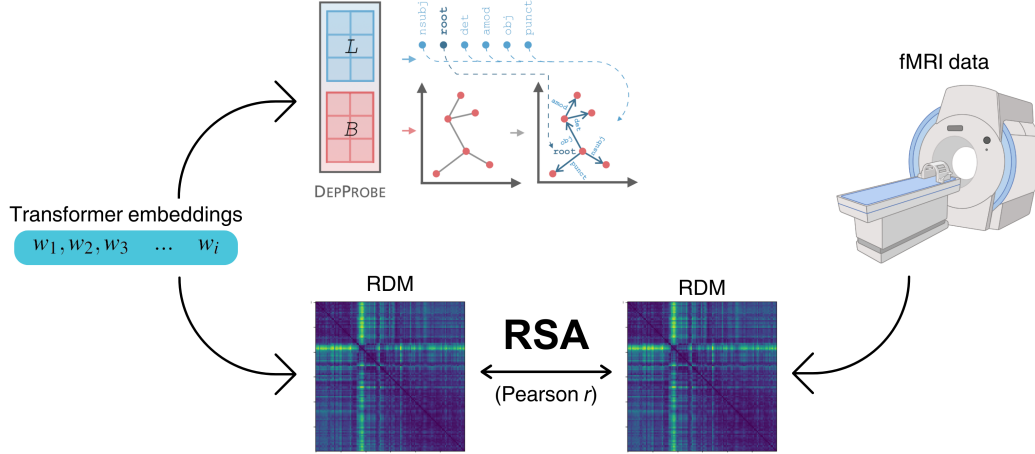


Figure 2: Schematic overview of the procedure. Participants read sentences while their brain signals were recorded with fMRI. The same sentences were fed through the five transformers, whose embeddings were then used with DepProbe to infer the dependency information represented across layers. The dependency information is then compared with the gold parses to arrive at the LAS for each layer. Next, we performed the RSA. The embeddings from each layer of the transformers and the brain signals were represented in cosine RDMs. Finally, the RDMs were compared with one another using the Pearson correlation coefficient. We could then compare the Pearson r with the accuracy of the represented dependency information in terms of LAS across layers. Adapted with permission from Müller-Eberstein et al. (2022).

et al., 2022) $(-50, -36, -2)$. Correct placement of the mask was visually verified using the Harvard-Oxford cortical and subcortical structural atlases (Frazier et al., 2005). Then, the BOLD signals were sliced using a standard delay of 4 seconds and averaged for each sentence with 4-7 TRs per sentence depending on the sentence length.

The extracted embeddings from all five transformer models were permuted to match the stimuli presentation order of each participant. Next, we performed the RSA. For each model layer from 4-11 and each participant, a representational dissimilarity matrix (RDM) was created. This matrix encodes the cosine distance between stimuli. RDMs were created for the BOLD signals of each participant and each set of model embeddings. RDMs were created for the representations of all the sentence stimuli and each group of sentence stimuli split by their LBC value. Since the LBC value could range from 1—4, this resulted in the creation of four RDMs for each model layer and subject. Next, the Pearson correlation was calculated between the RDMs of each model and each subject. This gave us the RS score, or, brain score, which was then averaged over all participants.

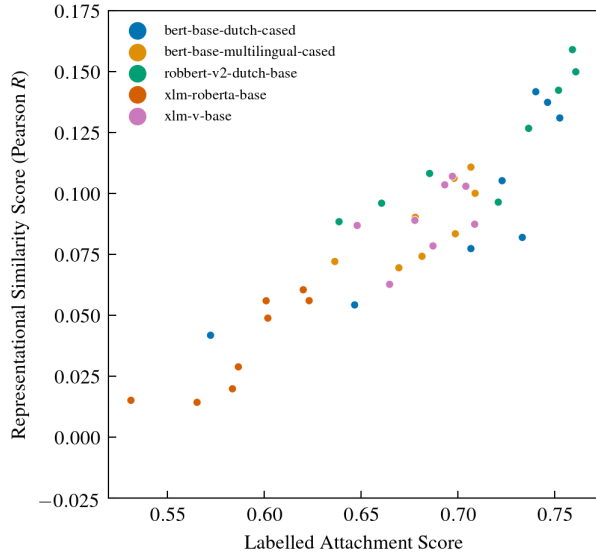
To confirm that our results can be ascribed to the dependency representations in the layers of the model, we created a control condition similar to the method used in Kauf et al. (2023). In this condition, the models were fed perturbed versions of the sentences that did not contain meaningful segments of three or more consecutive words. In this way, most of the meaningful dependency information was abolished and the remaining representational similarity can be ascribed to word meaning rather than syntactic structure.

Results

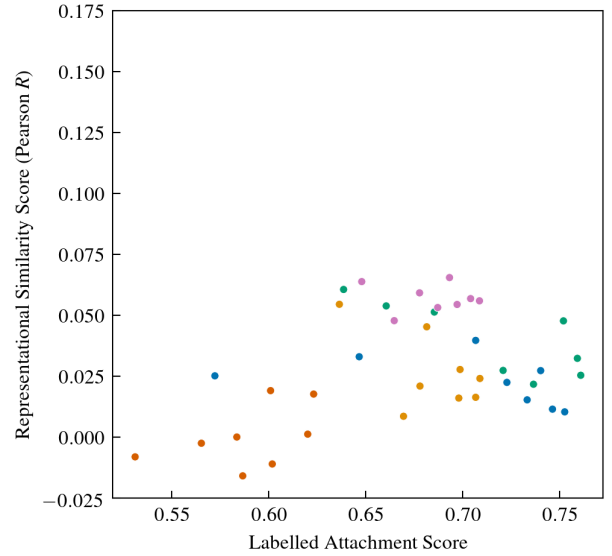
Model Layers That Accurately Represent Dependencies Are More Brain-like

The results from the RSA show that transformers that accurately represent dependency information yield higher RS scores, see Figure 3a. Results from the mixed model with the language models as a random intercept reveal a significant positive relationship between LSA and RS score ($b = 0.55$, $t = 11.76$, $SE = 0.05$, $p < .001$, 95% CI = $[0.46 - 0.64]$). This effect dissipates in the control condition ($b = -0.09$, $t = -1.91$, $SE = 0.05$, $p = .06$) thus this result cannot be ascribed to word meaning alone, see Figure 3b.

A similar pattern can be observed by comparing Figure 4a with Figure 4b. Here, we see that representational similarity across layers follows largely the same pattern as the LAS across layers. The RS scores from the control condition do not follow the same pattern, see Figure 4c. A paired-samples t-test was conducted to compare the RS scores for each layer 4-11 of each model between the condition in which the sentences were unperturbed and the control condition. There was a significant difference in the RS scores in the unperturbed condition ($M = 0.086$, $SD = 0.036$) and control condition ($M = 0.029$, $SD = 0.022$), $t(39) = 10.44$, $p < .001$. The magnitude of the differences in the means (mean difference = 0.057 , 95% CI: $[0.046 - 0.068]$) was very large (Cohen’s $d = 1.84$). Thus, we can conclude that perturbing the word order of sentences significantly influences RS scores which implies that word meaning alone cannot account for the brain alignment of the transformer models.



(a) Plot of the LAS and RS scores for the unperturbed condition



(b) Plot of the LAS and RS scores in the control condition

Figure 3: The LAS and RS scores for each layer from 4-11 of each model. In the unperturbed condition, there exists a positive relationship between LAS and RS scores while this pattern is absent in the control condition. This shows that model layers that accurately represent dependency information are more similar to the brain

Effect of Mono- Versus Multilingual Models

Examining Figures 3 and 4, Dutch monolingual models (BERTje and RobBERT) appear to surpass multilingual models (XLM-r, XLM-v, mBERT) in terms of both LAS and RS scores. Welch’s T-tests revealed significant LAS and RS score differences between monolingual (LAS: $M = 0.71$, RS: $M = 0.11$) and multilingual models (LAS: $M = 0.65$, RS: $M = 0.07$), with large effect sizes (Cohen’s d : LAS = 1.05, RS = 1.16, $p < .01$). However, upon closer expectation this difference seems to be driven primarily by one outlying model, XLM-R, see Figures 3a and 3b. Excluding XLM-R data nullifies these differences (LAS: $p = .12$; RS: $p = .051$), suggesting the initial performance gap is primarily attributed to XLM-R. Thus, we cannot conclude that monolingual models perform better on the metrics of LAS and RS score, but rather that model performance may vary, regardless of how many languages they are trained on.

Syntactic Complexity Influences Both Brain Similarity And Dependency Accuracy But Not Their Relationship

The MOUS dataset includes sentences of different levels of syntactic complexity, indexed by their Left-Branching Complexity (LBC) score. Because this may influence both brain alignment and the accuracy of dependency representation, we investigated whether syntactic complexity affected the RS score, the LAS, and the relationship between the LAS and RS scores, see Figure 5. For each LBC value, we built a linear mixed model with the LLMs as a random intercept to investigate the relationship between LAS and RS scores. The results of this analysis revealed that this relationship is signifi-

icant and positive for all LBC values: LBC 1, ($b = 0.80$, $t = 7.51$, $SE = 0.11$, $p < .001$, 95% CI = [0.59 - 1.00], LBC 2 ($b = 0.30$, $t = 2.19$, $SE = 0.13$, $p < .05$, 95% CI = [0.02 - 0.56], LBC 3 ($b = 0.73$, $t = 6.98$, $SE = 0.10$, $p < .001$, 95% CI = [0.52 - 0.99], LBC 4 ($b = 1.20$, $t = 3.92$, $SE = 0.31$, $p < .001$, 95% CI = [0.43 - 1.81]).

To investigate whether syntactic complexity influences RS scores, we performed a Kruskal-Wallis H test to evaluate differences in RS scores across the four LBC values. Results indicated a significant difference in RS scores across the groups, $H(3) = 94.07$, $p < .001$. Post-hoc comparisons using a Dunn test with a Bonferroni correction for multiple comparisons revealed that the RS scores of LBC 1 were significantly larger than those of LBC 2 ($Z = 3.62$, $p < .001$), and no significant difference was found between the RS scores from LBC 3 and LBC 4 ($Z = -0.91$, $p = 1$). For all other pair-wise comparisons, it holds that larger LBC values lead to higher RS scores ($Z > 4$, $p < .001$).

To investigate whether syntactic complexity influences LAS, we again employed a Kruskal-Wallis H test to evaluate differences in LAS across the four LBC values. Results indicated a significant difference in LAS across the groups, $H(3) = 45.45$, $p < .001$. These results indicate that overall syntactic complexity influences LAS. Post-hoc comparisons using a Dunn test with a Bonferroni correction for multiple comparisons revealed that the LAS of adjacent LBC values did not differ significantly. However, for the other pair-wise comparisons it holds that higher LBC values lead to lower LAS ($Z < -4$, $p < .001$).

Thus, we can conclude that syntactic complexity does not alter the relationship between the accuracy of dependency

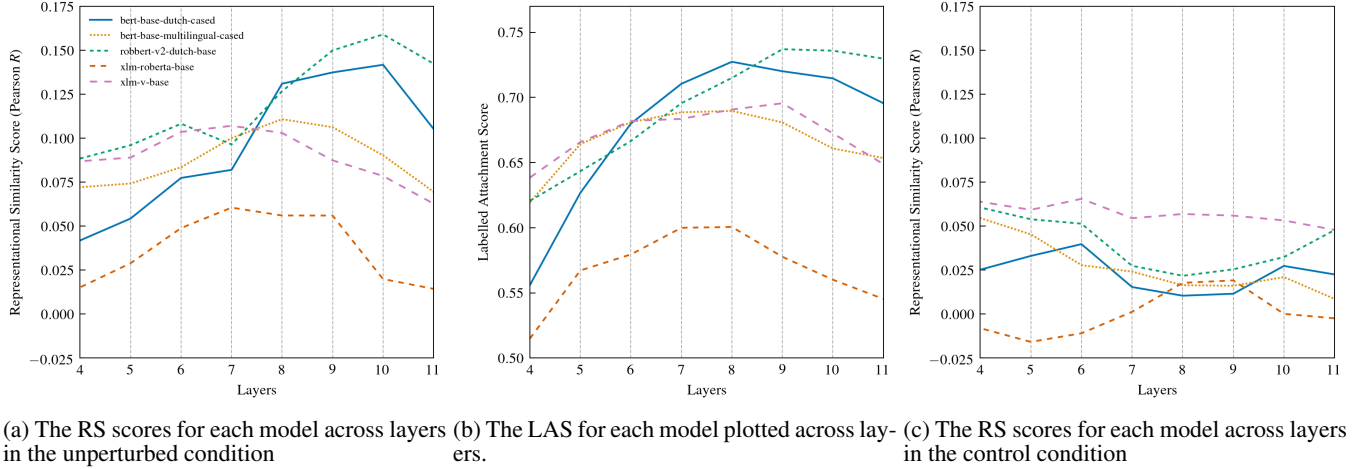


Figure 4: Comparison between the RS scores across all model layers in the unperturbed (4a) and control condition (4b), and the LAS across all model layers (4c). The LAS and RS scores in the unperturbed condition follow largely the same pattern, while this pattern is absent in the control condition.

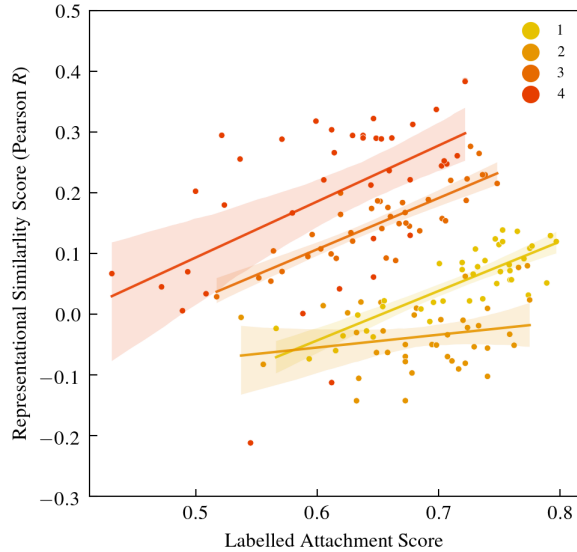


Figure 5: The LAS and RS scores for each layer from 4-11 of each model, with the scores divided over 4 groups representing the different levels of syntactic complexity. All levels of syntactic complexity follow the same pattern seen in 3. In addition, two patterns seem to emerge which indicate that syntactic complexity increases RS score, but lowers LAS.

representations and brain similarity observed previously. Yet, we found a surprising pattern that shows that syntactic complexity lowers dependency accuracy while increasing brain similarity.

Discussion

Firstly, our findings suggest a positive relationship between the accuracy with which transformer models represent dependency structures and their representational similarity to

the brain. Secondly, if we take all monolingual models and compare them to all multilingual models, the results indicate that monolingual models generally outperform multilingual models in terms of representing dependency information and, thus, following the confirmation of our first hypothesis, brain similarity. However, when we removed the worst-performing multilingual model, XLM-R, from the analysis, this effect dissipated. This suggests that accurately representing dependency information and brain similarity scores are more likely to be due to individual models than whether they are trained on data from one or multiple languages. Thirdly, we have shown that the relationship between LAS and RS scores persists when the data when grouped by syntactic complexity as measured by LBC. Additionally, two weaker but noticeable effects showed up in the analysis. These effects indicated that when syntactic complexity increases, the representations of the model layers become more brain-like and, in contrast, the accuracy of the dependency representations decreases. Therefore, there is a dissociation between how the LBC value of a sentence influences the LAS, and how it influences the RS scores.

Accuracy of Dependency Representations and Brain Score

Our results show that there exists a strong, positive relationship between the accuracy with which a model layer represents dependencies and their similarity to the brain. This is in line with earlier research by Murphy et al. (2012) and Abnar et al. (2017). Specifically, we found a positive relationship between the LAS of model layers and their representational similarity to the LpMTG, a brain area associated with syntactic processing. (Hagoort & Indefrey, 2014; Tyler et al., 2013). This finding suggests that for the purpose of modelling human language processing, it is beneficial to choose models that accurately represent dependency informa-

tion. This contrasts with earlier findings by Kauf et al. (2023) which suggested that semantic information was more important for brain similarity. Note, however, that the findings of the present study are confined to the LpMTG which has been shown to display higher activity when syntactic complexity goes up. Since the sentences used in the MOUS are particularly complex on a syntactic level, this may emphasize the importance of dependency representations for brain similarity. Many comparative studies between language models and brain data are focused on the language network as a whole or larger areas known to perform multiple functions. Many comparative studies between language models and brain data are focused on the language network as a whole which may obscure more fine-grained divisions of labour within the network. Future research can attempt to map particular aspects of language models on more specific ROIs which could lead to a more detailed understanding of how various functional representations within LLMs relate to the brain. Furthermore, the present study used RSA to investigate brain similarity while the study by Kauf and colleagues used regression. To our knowledge, there are no studies directly comparing these methods and how they influence the results of comparative cognitive AI studies. It would be interesting to see whether choosing one of either method significantly impacts the results of comparative cognitive AI studies.

Performance of Monolingual and Multilingual Models

From our findings, we cannot conclude that monolingual models outperform multilingual models in terms of LAS and brain score although the best-performing models in the present study were monolingual. Future research could investigate whether this pattern persists when comparing a more diverse set of mono- and multilingual models. It is important to note that, in the present study, we have only used non-finetuned transformer models. It may be the case that finetuning will further abolish the difference in both LAS and brain similarity between monolingual and multilingual models.

Syntactic Complexity

Our results suggest that, for each model, there exists an interesting dissociation between the relationship between LAS and syntactic complexity, which is negative, and the relationship between RS scores and syntactic complexity, which is positive. Yet, when we group LAS and RS scores by their LBC values, we can still see the positive relationship between LAS and RS scores. The fact that LAS and syntactic complexity have a negative relationship could be explained by the fact that sentences that are syntactically more complex are simply more difficult to parse. Indeed, some of the sentences present in the MOUS are so difficult to parse that even native speakers might have difficulty identifying dependencies on their first reading. An explanation for the positive relationship between RS score and syntactic complexity may be found by considering the function of the LpMTG. Earlier research by

Uddén et al. (2022) has indicated that the LpMTG becomes more active when a sentence is syntactically more complex. Thus, for syntactically more intricate sentences, we would expect a stronger signal which, in turn, would mean a higher signal-to-noise ratio (Welvaert & Rosseel, 2013). Therefore, it may be the case that the higher RS scores associated with higher syntactic complexity are simply due to an increase in detectable signal.

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